3.9 Internal audits by the company and external audits by the Administration of the ship's
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3.9 Internal audits by the company and external audits by the Administration of the ship's
safety management system should verify that the established procedures 1 are followed and
are consistent with the safety strategy referred to in these recommendations.
3.10 The company should establish criteria to manage additional risks involved during
simultaneous operations (SIMOPS) where one of the operations includes an enclosed space
entry. The criteria should consider the assessment of personnel and resources availabl e in
case of an emergency following an enclosed space entry.
1 Established procedures include resolution A.1050(27) as amended, the IMSBC Code, the BLU Code, the
IMDG Code, the IBC Code, the IGC Code, Recommendations on the safe use of pesticides in ships
applicable to the fumigation of cargo holds (MSC.1/Circ.1264), Guidelines on tank entry for tankers using
nitrogen as an inerting medium (MSC.1/Circ.1401) and other accepted standards and industry codes of
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nitrogen as an inerting medium (MSC.1/Circ.1401) and other accepted standards and industry codes of
practice, as relevant.
RESOLUTION MSC.581(110) (adopted on 27 June 2025)
REVISED RECOMMENDATIONS FOR ENTERING
ENCLOSED SPACES ABOARD SHIPS
MSC 110/21/Add.3
Annex 28, page 6
I:\MSC\110\MSC 110-21-Add.3.docx
3.11 The company should ensure that single person entry into an enclosed space is
not permitted.
4 IDENTIFICATION OF THE HAZARDS AND ASSESSMENT OF RISK
4.1 The company should ensure that a risk assessment is conducted to identify all
enclosed spaces on board the ship and that the identified enclosed spaces are recorded in the
Enclosed Space Register , which should be maintained on board the ship as well as ashore.
This Enclosed Space Register and risk assessment should be kept up to date as appropriate
to ensure its continued validity, particularly after loading and during the carriage of cargoes
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to ensure its continued validity, particularly after loading and during the carriage of cargoes
which may a dversely affect the safety of the atmosphere within a space. A reassessment
should also be made when the contents of the space change, such as in case a ship stores
treated sewage or grey water temporarily in its ballast water tanks. The company should
consider the use of appropriate technology to assist in the hazard identification and mitigation.
The Enclosed Space Register and risk assessment should form the basis of the development
of the enclosed space emergency response plan (app endix 1). It is recommended that the
enclosed space emergency response plan be reviewed after each drill so that its effectiveness
can be assessed and, if necessary, improvements made.
4.2 As entries for enclosed spaces other than cargo spaces may be different, there is a
need to record information for the benefit of personnel involved. Every ship should have an

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need to record information for the benefit of personnel involved. Every ship should have an Enclosed Space Register, the information within which may form the basis of a r isk assessment. The Enclosed Space Register should contain:
.1 physical layout of the space and access and egress points, including of connected spaces, if any;
.2 physical hazards in the space, e.g. vertical ladders, unguarded openings, poor lighting, wet or slippery conditions, excessive heat;
.3 connection to adjacent spaces;
.4 specific hazards within the space, for example, the effect of ballast water treatment method on the atmosphere within ballast tanks;
.5 if used, information related to additional technology, helping to determine enclosed space condition;
.6 information related to fixed and portable ventilation systems including equipment and where the equipment is stored;
.7 estimated time taken to achieve the air changes for safe entry, using forced or natural ventilation;

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.7 estimated time taken to achieve the air changes for safe entry, using forced or natural ventilation;
.8 lighting and means for temporary lighting including intrinsically safe lighting where appropriate;
.9 means for atmosphere testing;
.10 any pertinent information that would assist the risk assessment process;
.11 locking and "Safe to enter"/"Unsafe to enter" signage arrangements; and
.12 the equipment necessary to facilitate emergency rescue from the space.

4.3 In order to ensure safety from the outset, entry and occupancy, a competent person should make an assessment of any potential hazards in the space to be entered, taking into account the characteristics of the previous and current cargo carried with Safety Data Sheets
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account the characteristics of the previous and current cargo carried with Safety Data Sheets
(SDS) and cargo information, ventilation of the space, coating of the space and other relevant factors. Opening of an enclosed space should be subject to a risk assessment, taking into consideration the potential for release of dangerous ga ses or potential to create an explosive atmosphere: the competent person's assessment should determine the potential for the presence of an oxygen -deficient, oxygen -enriched, flammable or toxic atmosphere, which

includes carbon monoxide (CO) and carbon dioxide (CO2) as well as other toxic or asphyxiant gases. The competent person should bear in mind that the ventilation procedures for an adjacent or connected space may be different from those for the enclosed space itself. The details of the assessment should be recorded in a standard format, and maintained on board the ship.

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details of the assessment should be recorded in a standard format, and maintained on board the ship.

4.4 The nature of the hazards that may be present should be fully understood by those on board the ship, both crew and shore-based personnel, and those ashore managing the ship.

4.5 The procedures to be followed for testing the atmosphere in the space and for entry should be decided on the basis of the assessment. These will depend on whether the assessment shows that:

.1 there is minimal risk to the health or life of personnel entering the space; or

.2 there is no immediate risk to health or life, but a risk could arise during the course of work in the space; or

.3 a risk to health or life is identified.

4.6 Where the assessment indicates minimal risk to health or life or potential for a risk to arise during the course of work in the space, the precautions described in sections 5 to 8 and 10 should be followed, as appropriate.

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arise during the course of work in the space, the precautions described in sections 5 to 8 and 10 should be followed, as appropriate.

4.7 Where the assessment identifies a risk to life or health, if entry is to be made, the additional precautions specified in section 9 should also be followed.

4.8 Throughout the assessment process, there should be an assumption that the space to be entered is considered as hazardous until positively proved to be safe for entry.

5 AUTHORIZATION OF ENTRY

5.1 No person should open or enter an enclosed space unless authorized by the master or a responsible person nominated by the master and the appropriate safety procedures laid down for the particular ship have been followed.

5.2 Entry into enclosed spaces should be planned and the use of an entry permit system, which may include the use of a checklist, should always be used where the assessment

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which may include the use of a checklist, should always be used where the assessment identifies any risk to health or life. Prior to the entry of a space, an enclosed space entry permit should be issued by the master or the responsible person nominated by the master, and complied with at all times by the competent person included in the work especially those who enter the space. An example of the enclosed space entry permit is provided in appendix 2. The validity of the permit should be specified based on the risk assessment and should never be longer than eight hours.

5.3 Additional permits may be required depending upon the ships safety management system (SMS) and the activity being planned within the enclosed space. These permits may
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system (SMS) and the activity being planned within the enclosed space. These permits may include energy isolation permits, electrical isolation permit, hot work permit, working at height permit or the completion of documents such as Job Hazard Analysis.
5.4 Shore personnel should be provided with information on the hazards associated with enclosed spaces that may be present aboard the ship.
6 GENERAL PRECAUTIONS
6.1 Access to enclosed spaces should be carefully managed on ships where shore personnel regularly enter cargo spaces to load and discharge cargoes, especially on bulk carriers and general cargo ships. These recommendations are complementary to the requirements of the International Ship and Port Facility Security (ISPS) Code. It is recommended that a basic schematic plan of the ships spaces should be placed at the gangway or at other access points of the ship clearly indicating which spaces are safe to enter,
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gangway or at other access points of the ship clearly indicating which spaces are safe to enter, and which spaces are not. The plan should outline the enclosed space entry procedures that should be complied with by both ship and shore personnel before any entry is permitted. The use of simple safe to enter and unsafe to enter symbols is recommended. The plan should be time-stamped and dated and amended as necessary by the responsible person. This may also be beneficial on ships where large numbers of shore personnel are bei ng employed on the ship, such as in dry dock or ship repair facilities. An example of a ships schematic plan is contained in appendix 3.
6.2 Before any personnel are authorized to enter any cargo space containing any cargo, the competent person should carry out a risk assessment of the cargo; identification of the physical characteristics of the cargo space(s) concerned; and the operations to be carried out, and the responsible person should:

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physical characteristics of the cargo space(s) concerned; and the operations to be carried out,